

GROSSE ILE, MI, USA

WMO: 725373

Lat: 42.099N

Lon: 83.161W

Elev: 179

StdP: 99.19

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-12.5	-20.9	0.6	-14.3	-17.6	0.8	-11.4	11.1	-4.7	10.1	-4.7	3.5	240	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	31.6	23.5	29.8	23.0	28.0	22.1	25.4	29.3	24.5	28.1	23.7	27.1	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.3	27.6	23.0	18.1	26.6	22.5	17.6	26.2	78.5	29.2	74.6	28.1	71.8	27.4	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.3	7.5	-18.2	34.3	3.7	2.0	-20.9	35.7	-23.1	36.9	-25.2	38.0	-27.9	39.5		
			-18.7	26.9	3.5	1.2	-21.2	27.7	-23.3	28.4	-25.2	29.0	-27.8	29.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-3.0	-1.9	2.6	9.2	15.6	20.9	23.4	22.6	19.0	12.1	5.6	0.1
	DBStd	10.34	5.72	5.30	5.38	4.46	4.23	3.64	3.01	2.71	3.68	4.33	4.57	4.96
	HDD10.0	1519	402	332	236	64	4	0	0	0	0	27	144	308
	HDD18.3	3294	661	566	487	274	106	16	1	2	36	198	382	566
	CDD10.0	1738	0	0	8	42	179	326	416	392	269	92	13	1
	CDD18.3	472	0	0	0	2	22	93	158	136	55	6	0	0
	CDH23.3	3177	0	0	2	19	143	612	1247	874	263	17	0	0
	CDH26.7	793	0	0	0	2	29	162	361	192	46	2	0	0
(14) Wind	WSAvg	3.4	4.0	3.8	3.7	3.9	3.2	2.9	2.7	2.6	2.8	3.4	3.7	3.8
(15) Precipitation	PrecAvg	854	58	53	59	78	92	84	84	79	79	69	62	58
	PrecMax	1178	102	126	123	138	198	145	152	152	156	175	146	114
	PrecMin	658	21	18	18	17	34	33	34	13	20	9	15	20
	PrecStd	117	23	26	28	28	41	33	31	36	42	32	28	22
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	13.7	21.0	26.3	29.1	32.6	34.0	32.6	30.2	25.9	18.1	14.0
		MCWB	11.0	10.4	15.8	16.9	19.8	22.7	23.7	24.1	23.3	20.1	13.7	12.1
	2%	DB	8.0	8.8	17.0	22.1	27.3	30.2	32.2	30.6	27.7	22.6	16.0	11.1
		MCWB	6.6	6.8	12.7	15.1	19.8	22.3	23.9	23.6	21.8	18.4	12.5	9.8
	5%	DB	6.3	6.5	12.7	18.0	24.2	28.3	30.1	28.6	26.3	21.0	13.7	7.9
		MCWB	4.8	4.3	9.4	12.8	18.4	21.9	23.4	23.1	21.0	17.3	11.0	6.5
	10%	DB	4.0	4.1	10.0	16.2	22.4	27.2	28.5	27.4	24.5	18.7	12.3	6.4
		MCWB	2.7	2.6	7.4	11.5	17.4	21.0	22.8	22.1	20.3	15.7	10.0	4.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.4	16.4	18.4	23.0	25.2	27.3	26.7	25.0	21.5	15.0	12.5
		MCDB	11.7	12.5	20.1	23.7	27.1	29.8	31.1	30.8	28.6	24.5	17.0	13.7
	2%	WB	7.5	7.5	13.0	16.2	21.2	24.1	25.7	25.1	23.5	19.7	13.0	9.5
		MCDB	7.9	8.8	15.5	19.8	25.2	28.0	29.5	28.7	26.5	21.8	15.5	10.9
	5%	WB	5.1	4.7	10.2	14.3	19.7	23.3	24.7	24.1	22.4	17.7	11.6	7.1
		MCDB	6.5	6.3	12.6	18.0	23.3	27.0	28.5	27.3	25.1	19.6	13.3	7.9
	10%	WB	3.1	3.1	7.6	12.2	18.4	22.2	23.8	23.3	21.3	16.2	10.0	5.0
		MCDB	4.1	4.4	9.4	15.1	21.4	25.6	27.4	26.3	23.9	18.3	11.9	6.6
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.8	7.9	9.1	9.1	8.7	8.5	8.0	8.2	7.9	6.7	5.4
		MCDBR	7.8	9.3	11.9	12.6	11.3	9.9	9.5	9.0	9.1	9.6	8.8	7.4
	5% WB	MCWBR	7.0	7.8	8.5	7.9	6.4	5.0	5.0	4.9	5.3	6.2	6.9	6.6
		MCWBR	7.7	9.1	11.4	11.8	10.1	8.9	8.5	7.8	7.8	8.3	8.4	7.4
(40) Clear-Sky Solar Irradiance	taub		0.304	0.337	0.356	0.402	0.436	0.451	0.442	0.435	0.391	0.362	0.321	0.316
		taud	2.404	2.292	2.339	2.241	2.191	2.197	2.247	2.256	2.351	2.411	2.492	2.414
	Ebn at Noon		840	858	888	867	842	828	832	826	840	822	817	789
		Edh at Noon	79	104	112	133	143	143	135	130	110	92	73	73
(44) All-Sky Solar Radiation	RadAvg		1.49	2.37	3.49	4.62	5.41	6.07	6.09	5.30	4.28	2.80	1.75	1.22
	RadStd		0.15	0.29	0.34	0.39	0.46	0.39	0.32	0.20	0.37	0.23	0.23	0.11

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		-0.66	N/A	N/A	N/A	N/A	N/A	+186	+190	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A

Nomenclature: See separate page